

6-1 · Division Expressions with Fractions and Whole Numbers

Connect & apply · 8 tasks · Prepares for 6.NS.A.1

Name: _____ Date: _____

Show your thinking. Label units. Use a sketch, words, or equations. Take the time you need.

1. FOUNDATION 1

Compute $8.4 \div 4$.

2. FOUNDATION 2

What is the reciprocal of 5?

3. FOUNDATION 3

Compute $12 \times \frac{2}{3}$.

4. FOUNDATION 4

Write $\frac{15}{3}$ as a division expression and evaluate it.

One strategy I used: _____

My next step: ☐ Explain it to someone ☐ Try another example ☐ Ask for help

6-1 · Division Expressions with Fractions and Whole Numbers

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Name: _____ Date: _____

Show your thinking. Label units. Use a sketch, words, or equations. Take the time you need.

5. CHECK A

$4.8 \div 2$? How many fourths are in 3? Reciprocal of 5?

6. CHECK B

$6.4 \div 2$? How many thirds are in 4? Reciprocal of 8?

7. CHECK C

$9.6 \div 4$? How many halves are in 5? Reciprocal of $\frac{1}{6}$?

8. TRANSFER

A ribbon is 6 feet long. How many $\frac{3}{4}$ -foot pieces can be cut from it?

One strategy I used: _____

My next step: ☐ Explain it to someone ☐ Try another example ☐ Ask for help

6-2 · Division Expressions with Fractions and Mixed Numbers

Connect & apply · 8 tasks · Prepares for 6.NS.A.1

Name: _____ Date: _____

Show your thinking. Label units. Use a sketch, words, or equations. Take the time you need.

1. FOUNDATION 1

Write $2\frac{3}{4}$ as an improper fraction.

2. FOUNDATION 2

Compute $\frac{2}{3} \times \frac{3}{5}$.

3. FOUNDATION 3

Find the reciprocal of $\frac{4}{7}$.

4. FOUNDATION 4

Simplify $\frac{18}{24}$.

One strategy I used: _____

My next step: ☐ Explain it to someone ☐ Try another example ☐ Ask for help

6-2 · Division Expressions with Fractions and Mixed Numbers

Connect & apply · 8 tasks · Prepares for 6.NS.A.1

Name: _____ Date: _____

Show your thinking. Label units. Use a sketch, words, or equations. Take the time you need.

5. CHECK A

2 3/4 as an improper fraction? Reciprocal of 3/5?
2/3 × 5/4?

6. CHECK B

3 2/5 as an improper fraction? Reciprocal of 4/7?
3/4 × 2/5?

7. CHECK C

1 5/6 as an improper fraction? Reciprocal of 9/2?
5/6 × 3/10?

8. TRANSFER

How many 1 1/2-cup servings are in 6 cups?

One strategy I used: _____

My next step: ☐ Explain it to someone ☐ Try another example ☐ Ask for help

6-3 · Explore Numerical Expressions with Exponents

Connect & apply · 8 tasks · Prepares for 6.EE.A.1

Name: _____ Date: _____

Show your thinking. Label units. Use a sketch, words, or equations. Take the time you need.

1. FOUNDATION 1

Compute $3 \times 3 \times 3 \times 3$.

2. FOUNDATION 2

Evaluate 5^2 .

3. FOUNDATION 3

Evaluate $10 \times 10 \times 10$.

4. FOUNDATION 4

How many factors of 2 appear in $2 \times 2 \times 2 \times 2 \times 2$?

One strategy I used: _____

My next step: ☐ Explain it to someone ☐ Try another example ☐ Ask for help

Connect & apply · 8 tasks · Prepares for 6.EE.A.1

Name: _____ Date: _____

5. CHECK A

My next step: ☐ Explain it to someone ☐ Try another example ☐ Ask for help

6-4 · Write and Evaluate Numerical Expressions with Exponents

Connect & apply · 8 tasks · Prepares for 6.EE.A.2.c

Name: _____ Date: _____

Show your thinking. Label units. Use a sketch, words, or equations. Take the time you need.

1. FOUNDATION 1

Evaluate 4^3 .

2. FOUNDATION 2

Evaluate $2 + 3^2 \times 4$.

3. FOUNDATION 3

Evaluate $(8 + 4)/3$.

4. FOUNDATION 4

Write an expression for 7 less than the product of 3 and 5.

One strategy I used: _____

My next step: ☐ Explain it to someone ☐ Try another example ☐ Ask for help

6-4 · Write and Evaluate Numerical Expressions with Exponents

Connect & apply · 8 tasks · Prepares for 6.EE.A.2.c

Name: _____ Date: _____

Show your thinking. Label units. Use a sketch, words, or equations. Take the time you need.

5. CHECK A

$4 + 6 \div 2$? $3 + 2^2 \times 5$? Which operation comes first?

6. CHECK B

$8 + 12 \div 4$? $5 + 3^2 \times 2$? Which operation comes first?

7. CHECK C

$20 - 6 \times 2$? $4 + 2^3 \div 4$? Which operation comes first?

8. TRANSFER

Evaluate $2 \times (3 + 4)^2 - 10$.

One strategy I used: _____

My next step: ☐ Explain it to someone ☐ Try another example ☐ Ask for help

Original supplemental practice · Lesson numbering follows this guide's district sequence · Discuss your reasoning; this is practice, not a speed test.

6-5 · Write and Evaluate Algebraic Expressions

Connect & apply · 8 tasks · Prepares for 6.EE.A.2

Name: _____ Date: _____

Show your thinking. Label units. Use a sketch, words, or equations. Take the time you need.

1. FOUNDATION 1

Evaluate $n + 8$ when $n = 6$.

2. FOUNDATION 2

Evaluate $2x + 3$ when $x = 5$.

3. FOUNDATION 3

Evaluate $4n$ when $n = 7$.

4. FOUNDATION 4

Write an expression for 9 more than a number n .

One strategy I used: _____

My next step: ☐ Explain it to someone ☐ Try another example ☐ Ask for help

6-5 · Write and Evaluate Algebraic Expressions

Connect & apply · 8 tasks · Prepares for 6.EE.A.2

Name: _____ Date: _____

Show your thinking. Label units. Use a sketch, words, or equations. Take the time you need.

5. CHECK A

Evaluate $4n + 3$ when $n = 5$. Write an expression for "7 more than a number."

6. CHECK B

Evaluate $3n + 8$ when $n = 6$. Write an expression for "5 less than a number."

7. CHECK C

Evaluate $2n - 7$ when $n = 10$. Write an expression for "the product of 6 and a number."

8. TRANSFER

Write an expression for the cost of n tickets at \$12 each plus a \$5 fee, then find the cost for 4 tickets.

One strategy I used: _____

My next step: ☐ Explain it to someone ☐ Try another example ☐ Ask for help

6-6 · Identify Equivalent Algebraic Expressions

Connect & apply · 8 tasks · Prepares for 6.EE.A.4

Name: _____ Date: _____

Show your thinking. Label units. Use a sketch, words, or equations. Take the time you need.

1. FOUNDATION 1

Simplify $5x + 2x$.

2. FOUNDATION 2

Evaluate $3x + 4$ when $x = 2$.

3. FOUNDATION 3

Evaluate $3(4 + 5)$ using distribution.

4. FOUNDATION 4

Compare $2(x + 3)$ and $2x + 6$ at $x = 4$. Give both values.

One strategy I used: _____

My next step: ☐ Explain it to someone ☐ Try another example ☐ Ask for help

6-6 · Identify Equivalent Algebraic Expressions

Connect & apply · 8 tasks · Prepares for 6.EE.A.4

Name: _____ Date: _____

Show your thinking. Label units. Use a sketch, words, or equations. Take the time you need.

5. CHECK A

$5x + 2x = ?$ Evaluate $2(x + 3)$ and $2x + 6$ at $x = 4$
— are they the same?

6. CHECK B

$8y + 3y = ?$ Evaluate $3(x + 2)$ and $3x + 6$ at $x = 5$
— are they the same?

7. CHECK C

$9a - 4a = ?$ Evaluate $4(x + 1)$ and $4x + 4$ at $x = 3$
— are they the same?

8. TRANSFER

Are $3(2x + 4)$ and $6x + 7$ equivalent? Justify.

One strategy I used: _____

My next step: ☐ Explain it to someone ☐ Try another example ☐ Ask for help

6-7 · Find Factors and Multiples

Connect & apply · 8 tasks · Prepares for 6.NS.B.4

Name: _____ Date: _____

Show your thinking. Label units. Use a sketch, words, or equations. Take the time you need.

1. FOUNDATION 1

Compute $84 \div 7$.

2. FOUNDATION 2

List every positive factor of 18.

3. FOUNDATION 3

List the first four positive multiples of 7.

4. FOUNDATION 4

Which of 2, 3, 5, 9, and 10 divide 135 evenly?

One strategy I used: _____

My next step: ☐ Explain it to someone ☐ Try another example ☐ Ask for help

6-7 · Find Factors and Multiples

Connect & apply · 8 tasks · Prepares for 6.NS.B.4

Name: _____ Date: _____

Show your thinking. Label units. Use a sketch, words, or equations. Take the time you need.

5. CHECK A

All the factors of 18? First five multiples of 6?
GCF of 18 and 24?

6. CHECK B

All the factors of 24? First five multiples of 8?
GCF of 24 and 36?

7. CHECK C

All the factors of 20? First five multiples of 5?
GCF of 20 and 30?

8. TRANSFER

You have 24 pencils and 36 erasers to split into identical kits with nothing left over. What is the largest number of kits?

One strategy I used: _____

My next step: ☐ Explain it to someone ☐ Try another example ☐ Ask for help

6-8 · Generate Equivalent Expressions

Connect & apply · 8 tasks · Prepares for 6.EE.A.3

Name: _____ Date: _____

Show your thinking. Label units. Use a sketch, words, or equations. Take the time you need.

1. FOUNDATION 1

Find the greatest common factor of 18 and 30.

2. FOUNDATION 2

Factor $12x + 20$ using the greatest common factor.

3. FOUNDATION 3

Simplify $4a + 3a + 2$.

4. FOUNDATION 4

Compute $6(10 + 4)$.

One strategy I used: _____

My next step: ☐ Explain it to someone ☐ Try another example ☐ Ask for help

6-8 · Generate Equivalent Expressions

Connect & apply · 8 tasks · Prepares for 6.EE.A.3

Name: _____ Date: _____

Show your thinking. Label units. Use a sketch, words, or equations. Take the time you need.

5. CHECK A

GCF of 18 and 24? Expand $3(x + 5)$. Factor $4x + 12$.

6. CHECK B

GCF of 20 and 30? Expand $5(x + 4)$. Factor $6x + 18$.

7. CHECK C

GCF of 12 and 16? Expand $4(x + 7)$. Factor $9x + 15$.

8. TRANSFER

Factor $24x + 36$ completely, then check by expanding.

One strategy I used: _____

My next step: ☐ Explain it to someone ☐ Try another example ☐ Ask for help